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STAT 632

Excercise 2 :

$$Y_i = \beta X_i + e_i \quad \text{— given}$$

Assuming $e_i \sim N(0, \sigma^2)$

a] For the Least Square Estimation.

$$Q = \Sigma(Y_i - \beta X_i)^2$$

Finding the derivative of Q

$$\frac{dQ}{d\beta} = -2 \Sigma X_i (Y_i - \beta X_i)$$

Setting the above equation to zero.

$$-2 \Sigma X_i (Y_i - \hat{\beta} X_i) = 0$$

$$\Sigma(X_i Y_i - \hat{\beta} X_i^2) = 0$$

$$\Sigma X_i Y_i = \Sigma \hat{\beta} X_i^2$$

$$\Sigma X_i Y_i = \hat{\beta} \Sigma X_i^2$$

$$\therefore \hat{\beta} = \frac{\Sigma X_i Y_i}{\Sigma X_i^2} = \frac{\Sigma_{i=1}^{n} x_i y_i}{\Sigma_{i=1}^{n} X_i^2}$$

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b] Solution:

$$E(\hat{\beta}|X) = E\left\{\frac{\sum X_i Y_i}{\sum X_i^2} \middle| X\right\}$$

$$= \left\{\frac{\sum X_i \beta X_i}{\sum X_i^2}\right\}$$

$$= \left\{\frac{\beta \cancel{\sum X_i^2}}{\cancel{\sum X_i^2}}\right\}$$

$$E(\hat{\beta}) = \beta$$

$$\therefore E(\hat{\beta}) = \beta$$

This shows that $\hat{\beta} = \beta$ is unbiased.

c] Solution:

We will use the MSE as our estimator of σ^2, where

$$\sigma^2 = MSE = \frac{\sum e_i^2}{n-1} = \frac{SSE}{n-1}$$

Here, $E(\sigma^2) = E(MSE) = \sigma^2$

$$\sigma = S_e = \sqrt{\frac{\sum[Y_i - (\beta X_i)]^2}{n-1}} = \sqrt{\frac{\sum e_i^2}{n-1}}$$

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$$\sigma = \sqrt{\frac{SSE}{n-1}}$$

$$\sigma = \sqrt{MSE}$$

$$\sigma^2 = MSE$$